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| **Fiscal Year:** | FY 2022 | **Task Last Updated:** | FY 12/31/2021 |
| **PI Name:** | Hammond, Timothy M.D. | | |
| **Project Title:** | Fuel to Mars | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Microbiology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
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| **PI Organization Type:** | NON-PROFIT | **Phone:** | 9194285055 |
| **Organization Name:** | Institute of Medical Research/Department of Veterans Affairs | | |
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| **Comments:** | Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System. | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2018 Space Biology (ROSBio) NNH18ZTT001N-Artemis1 (EM1). App A: Orion (Artemis-1) (formerly Exploration Mission-1) |
| **Start Date:** | 03/18/2019 | **End Date:** | 03/17/2023 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 3 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | Zhang, Ye | **Contact Phone:** | 321-861-3253 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 03/17/2023 per NSSC information. | | |
| **Key Personnel Changes/Previous PI:** | January 2021 report: None. | | |
| **COI Name (Institution):** | Allen, Patricia M.S. ( Veterans Health Administration )<br>Birdsall, Holly M.D., Ph.D. ( Institute For Medical Research, Inc. )<br>Giaever, Guri Ph.D. ( University of British Columbia, Canada )<br>Nislow, Corey Ph.D. ( University of British Columbia, Canada ) | | |
| **Grant/Contract No.:** | 80NSSC19K0706 | | |
| **Performance Goal No.:** | | | |

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| **Performance Goal Text:** | NOTE: End date changed to 3/17/2022 per NSSC information (Ed., 5/18/21) |
| **Task Description:** | The primary research product this project will produce is identification of the insertion strains of Chlamydomonas reinhardtii, a single-cell green alga, which optimizes survival during exposure to the combined impacts of space radiation and microgravity in the cosmic space environment beyond the Van Allen Belts. The Chlamydomonas reinhardtii strains in a mapped insertional mutant library are a transformative tool allowing near genome wide analysis of the genes mediating survival advantage during stimuli such as cosmic radiation and microgravity. The secondary set of research products this project will produce are identification of the pathways mediating the gene changes, visualization of the pathway interactions, and identification of any environments giving similar gene pathway signatures in other model systems.<br>Our goal is to identify strains of Chlamydomonas reinhardtii with gene insertions, which convey the best survival advantage during the exposure to the combined impacts of space radiation and microgravity for future use as parent strains in hydrogen production initiatives. Hydrogen is the lead contender for fuel production for return space flight missions from orbital bodies beyond low Earth orbit. The metabolism of select microorganisms makes hydrogen fuel. The best-known and most efficient hydrogen producer is Chlamydomonas reinhardtii, which is poised for exploitation as an industrial biotechnology platform for biofuel production.<br><br>The type of investigation we propose is experiments that will be conducted aboard the Exploration Mission-1 (EM-1) [Note now called Artemis] lunar orbit flight, with asynchronous controls in Kennedy Space Center ISSES (International Space Station Environmental Simulators) Chambers. The proposed hardware is the NASA Biological Research in Canisters (BRIC)-100VC.<br><br>Hypothesis: The hypothesis to be tested is that the mapped insertional mutant library of Chlamydomonas reinhardtii will allow identification of the genes that mediate optimal radiation resistance, and imperviousness to microgravity, in the cosmic space environment beyond the Van Allen Belts.<br><br>The specific aims of the proposal are:<br><br>Specific Aim 1: To experimentally determine which of the thousands of mapped insertion mutant strains of Chlamydomonas reinhardtii have the best long-term survival advantage during combined space radiation and microgravity impacts in the cosmic space environment beyond the Van Allen Belts.<br><br>Specific Aim 2: To analyze the dataset of insertion strains with the best long-term survival advantage during combined space radiation and microgravity impacts to:<br><br>• Perform pathway analysis using gene set enrichment analysis (GSEA);<br><br>• Visualize the pathway interactions using Cytoscape/ClueGo;<br><br>• Make comparison to homologous gene sets in Chlamydomonas, Saccharomyces cerevisiae, and other model organism databases for any environments giving similar gene pathway signatures.<br><br>Although focused on gene pathways for survival for future applications in hydrogen fuel production, the data generated will provide genome level insights into the cancer risks of prolonged exposure to cosmic radiation and microgravity. The spectrum of cancer risk genes is known, and this study will parse the importance of risk genes, and the patterns of radiation risk change with exposure to the cosmic space environment beyond the Van Allen Belts. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The research impacts and Earth benefits include:<br>o The first opportunity to apply modern “omics” techniques to samples flown beyond the van Allen belts, and recover for ground-based analysis.<br><br>o The first biological validation of the utility of the Chlamydomonas deletion series for academic and commercial applications.<br><br>o Delineation of the production of the potential fuels, lipid and hydrogen, to optimize use in future space based and ground applications.<br><br>o Education of the next generation of space flight investigators.<br><br>o This proposal directly addresses the Space Biology Plan Element GL3: What are the key cellular and molecular systems necessary for biological organisms to thrive in space environment that can be discovered from 21st century bioinformatics tools?<br><br>o This application directly addresses three elements of the Human Research Roadmap on Risk of Radiation Carcinogenesis: Cancer 07: How can systems biology approaches be used to integrate research on the molecular, cellular, and tissue mechanisms of radiation damage to improve the prediction of the risk of cancer and to evaluate the effectiveness of countermeasures? How can epidemiology data and scaling factors support this approach? Cancer 08: What are the most effective medical or dietary countermeasures to mitigate cancer risks from exposure to SPE (solar particle events) and GCR (galactic cosmic rays)? What side effects should be tolerated versus mission risks? Can medical countermeasures that provide mitigation across multiple cancer types or across multiple radiation risk areas be identified? Cancer 09: Are there significant effects from other spaceflight factors that modify the carcinogenic risk from space radiation? |

**Task Progress:**

Maintenance of Chlamydomonas mapped insertional mutant library.
The Chlamydomonas mapped insertional mutant library was maintained on Tris- acetate-phosphate (TAP) agar plates and aliquots reseeded robotically onto fresh plates every 2 weeks. There have been no growth or contamination problems to date over the current 12 month reporting period.

Temperature sensitivity assessment.

Updated predicted temperature profile data from the Artemis program office led the Space Biology program office to ask for temperature sensitivity assessments of the phenotype of the Chlamydomonas during exposure to a new temperature profile as high as 49.4oC and as low as 3.8oC .

The hypothesis of this temperature assessment test was that exposure to 49.4oC will kill the Chlamydomonas but we will still be able to conduct the majority of the phenotyping assays listed in our proposal. Exposure to cold will retard growth, but will have minimal effect on viability as stock cultures of Chlamydomonas are routinely stored in the cold. Thus, the anticipated heat and/or cold exposures will have a minimal impact on the science in our experiments on EM-1.

The specific aims of the proposal were to measure programmed cell death, lipid and chlorophyll content, activity of photosystem pathways I and II, cell size distribution, and cytochrome f activity of Chlamydomonas reinhardtii after exposure to 49.4oC for 0, 1, and 36 hours and after exposure to 3.8oC for 8, 10, 20, and 28 days.

o 36 hours of heat exposure is included as the projected interval of exposure to 49.4oC between splashdown and Avionics shut down.

o Zero-hour exposure is included as a negative control.

o One hour exposure to 49.4oC is included as we anticipate that even this short exposure will kill all of the Chlamydomonas.

o 3.8oC was used as the cold exposure

The samples from the actual mission will be around 60 days old at the time of reentry and potential exposure to 49.4oC. Results from our Science Verification Test (SVT) showed that 60-day old colonies have greatly reduced viability, but lipid and chlorophyll A and B content, activity of photosystem pathways I and II, and cytochrome f activity could all still be measured. We are using fresh, viable cultures for this temperature assessment because spotting the Chlamydomonas on agar waiting for 60 days will not allow us to provide NASA with the requested assessment in a timely fashion.

Heat exposure

At time zero, three plates of Chlamydomonas were placed in an incubator set to 49.4oC and one plate was incubated at room temperature. One heat-exposed plate was assayed immediately after 1-hr exposure to 49.4oC. One heat-exposed plate transferred to room temperature after 1-hr exposure to 49.4 and maintained at room temperature for the remaining 35 hours before assay. The third heat-exposed plate remained at 49.4oC for the entire 36-hour period prior to assay. The last plate remained at room temperature for the entire 36-hour period.

Cold exposure

Four plates of Chlamydomonas were placed in a cold box, set to 3.8oC, and four plates were maintained at room temperature (RT). Both chilled and ambient temperature plates were exposed to the same cycling red and blue light coming on contemporaneously for 6 hours out of every 24-hour cycle. One plate from the cold box and one room temperature control plate were assayed after 8, 10, 20, or 28 days.

Assays proposed in grant application and validated in the Science Verification Test.

At the end of the exposure, the Chlamydomonas were harvested and assayed by spectrophotometry and then flow cytometry. These analyses for temperature sensitivity included all phenotyping parameters proposed as secondary readouts in the grant application and validated in the Science Verification Test.

Absorbance:

Chlamydomonas colonies were assayed by spectrophotometry for Chlorophyll A (absorbance at 436 nm); Chlorophyll B (absorbance at 472 nm); PS I and II (absorbance at 700 nm); Cytochrome f (absorbance at 554 nm); Protein content (absorption at 800 nm).

For each wavelength, background absorbance of agar alone (e.g., an area of the plate with no Chlamydomonas colony), was subtracted to arrive at the net absorbance. The net absorbance values were then normalized to the protein content as measured by absorbance at 800 nm.

Flow cytometry

Six colonies from each plate were harvested and re-suspended in annexin-binding buffer, which is a phosphate buffer with calcium. The aggregates were disrupted by vortexing and then filtered through nylon mesh to remove clumps that would plug the flow cytometer. The resuspended Chlamydomonas were aliquoted 100 microl per well into V-bottom microtiter plates and stained with:

• Alexa Fluor 488-annexin (5 microl/well) and propidium iodide (PI) (1 microl) to identify: % necrotic cells (PI positive, annexin negative); % early apoptotic cells (PI negative, annexin positive); % late apoptotic cells (PI positive, annexin positive); % live cells (PI negative and annexin negative).

• Nile Red to measure lipid content with triton X to solubilize the membrane.

We counted 2,000 Chlamydomonas from each sample. Background signal was estimated by appropriate no-dye controls. In every flow cytometry run, quality controls were performed on the instrument with 3 color fluorescent beads, followed by assay of 5 control tubes: no dyes, Alexa fluor 488 annexin binding alone, propidium iodide alone, Alexa fluor 488 annexin binding with propidium iodide, and Nile red alone. Flow cytometric and spectrophotometric data were evaluated by two-tailed Student's t-test comparing heat- or cold-exposed Chlamydomonas to the relevant room temperature control.

RESULTS

Effect of heat exposure (49.4oC) on photosynthesis proteins: Heat exposure to 49.4oC has minimal effect on the quantity of Chlorophyll A, Chlorophyll B, Cytochrome f, or Photosystem I and II in Chlamydomonas.

Effect of cold exposure (3.8oC) on photosynthesis proteins: Cold exposure to 3.8oC has minimal effect on the quantity of Chlorophyll A, Chlorophyll B, Cytochrome f, or Photosystem I and II in Chlamydomonas.

Flow cytometric assessment of viability, apoptosis, and lipid content:

• Approximately 15% of the Chlamydomonas in the control (room temperature) cultures are undergoing early apoptosis at 8 days. After 20-28 days at room temperature, approximately 5% of Chlamydomonas have progressed to late apoptosis or necrosis.

• After 20-28 days exposure to cold (3.8oC), 10-12% of the Chlamydomonas are in early apoptosis and 20-23% of the Chlamydomonas have progressed to late apoptosis or necrosis. Thus, viability is slightly, but significantly, diminished after prolonged exposure to cold temperatures.

• One hour of exposure to 49.4oC drives 70% of the Chlamydomonas to early apoptosis. 35 hours later, 86% of the Chlamydomonas are in late apoptosis.

• If maintained at 49.4oC for the entire 36-hr period, 83% of the Chlamydomonas are in late apoptosis.

• To remedy this issue, we will use the viability marker fluorescein diacetate at a concentration of 2.4 microM (1 microM/ml). We shall also change the fluorophore conjugated to annexin to an Alexa Fluor 555 conjugate so the apoptotic marker does not interference with the viability or cell death signals after color compensation. Gating on the viability marker, the apoptotic and necrotic markers can be run on the remaining live cells. As there are >400 million Chlamydomonas cells on each plate, sufficient sample for analysis should not be a problem.

• Lipid content is significantly greater in Chlamydomonas exposed to cold (3.8oC) for 8-10 days but by 20-28 days, the levels are no different from those in the RT control plates.

• Thus, exposure to 49.4oC for periods as short as one hour eliminates the uptake of Nile Red, suggesting that lipid content has been dissipated.

CONCLUSIONS

Analysis of the effects of temperature to the hot and cold limits of the expected thermal envelop allow several conclusions.

Primary outcome

1. The primary outcome of next generation sequencing to determine number of copies of each mapped deletion clones for analysis of survival pathways during exposure to space radiation should be entirely intact.

2. Heat to 49.4oC even for one-hour results in apoptotic death of the Chlamydomonas. This is actually a major experimental advantage as it will stop the primary outcome genomics part of the experiment, avoiding artefacts from the time of sample recovery and transport to Kennedy Space Center.

3. Cold, even prolonged to 30 days, has scant if any effect on the primary outcome of next generation sequencing to determine number of copies of each mapped deletion clones.

Secondary outcomes

1. Cold at the edge of the expected envelope, even prolonged at 8 to 10 days, has scant if any effect on the secondary outcomes of spectrophotometry analysis of Chlorophyll A, Chlorophyll B, PS I and II, Cytochrome f, and protein content.

2. Cold for 8-10 days has scant effect on the secondary outcomes of flow cytometry analysis of live cells, early apoptosis, late apoptosis, and necrosis, or Nile red content as an index of lipid content.

3. However, by 27 days in the cold at the very edge of the envelope, cold causes increased late apoptosis and necrosis. The live population of Chlamydomonas, while decreased, is still abundant and viable.

4. Heat, even 49.4oC for 36 hours, has scant effect on the secondary outcomes of spectrophotometry analysis of Chlorophyll A, Chlorophyll B, PS I and II, Cytochrome f, and protein content.

5. Heat, 49.4oC for 1 hour or more invalidates the secondary flow cytometry analysis of live cells, early apoptosis, late apoptosis, and necrosis, or Nile red content as an index of lipid content. This will be remedied by adding and gating on a viability marker.

Summary

1. Many secondary outcomes such as spectrophotometer assay of Chlorophyll A, Chlorophyll B, PS I and II, Cytochrome f, and protein content are robust to all envelope temperatures.

2. Some secondary outcomes, specifically lipid content, and programmed cell death, will be affected by prolonged cold or heat, but this will be remedied by addition of a viability marker.

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| **Bibliography Type:** | Description: (Last Updated: 03/31/2026) |
| **Abstracts for Journals and Proceedings** | Birdsall H, Hammond T, Nislow CE. "Effects of microgravity simulated with cell spinpods on Chlamydomonas." Session: Microbes 4 - Microgravity. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021.<br>Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021 |

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| **Abstracts for Journals and Proceedings** | Hammond TG, Nislow CE, Christov I, Allen PL, Birdsall HH. "Cell spinpods: a simple inexpensive rotating suspension culture device." Session: Enabling Technologies 2. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021.<br>Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021 |
| **Articles in Peer-reviewed Journals** | Hammond TG, Nislow C, Christov IC, Batuman V, Nagrani PP, Barazandeh M, Upadhyay R, Giaever G, Allen PL, Armbruster M, Raymond A, Birdsall HH. "Cell spinpods are a simple inexpensive suspension culture device to deliver fluid shear stress to renal proximal tubular cells." Sci Rep. 2021 Oct 29;11(1):21296.<br>https://doi.org/10.1038/s41598-021-00304-8 ; PMID: 34716334; PMCID: PMC8556299 , Oct-2021 |